

Lab 1: Introduction to Web Technologies and Understanding HTTP and HTTPS

Exercise 1: Creating a Simple Web Page

1. Creating a New HTML File:
2. Writing HTML Code:

Code Used:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Introduction to Web Technologies and Database
Security</title>
</head>
<body>
  <h1>Understanding Web Technologies</h1>

  <figure>
    <small></small>
  </figure>
```

<p> Web technologies are important because they make it possible
for

people and organizations to create, access, and share information

and services through the internet. They support websites, web
applications,

online communication, e-commerce, banking, education, and
many other digital services.

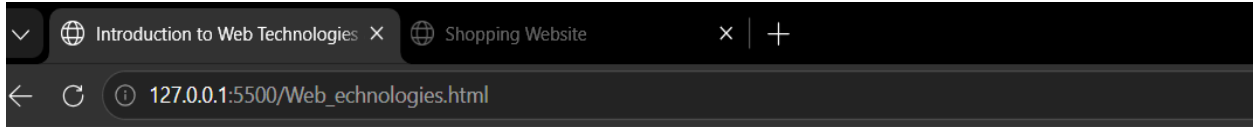
 Technologies such as HTML, CSS, JavaScript, HTTP, and
HTTPS work
together to ensure that

web pages are created, displayed, and communicated securely.

 As a result, web technologies play a major role in modern
communication,
business, education, and everyday digital activities.
</p>

<p>
Click here
</p>
</body>
</html>

```
Welcome Web_echnologies.html X
Web_echnologies.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Introduction to Web Technologies and Database Security</title>
7 </head>
8 <body>
9   <h1>Understanding Web Technologies</h1>
10
11   <figure>
12     <small></small>
14   </figure>
15
16 <p> Web technologies are important because they make it possible for <br>
17   people and organizations to create, access, and share information <br>
18   and services through the internet. They support websites, web applications,<br>
19   online communication, e-commerce, banking, education, and many other digital services.
20   <br> Technologies such as HTML, CSS, JavaScript, HTTP, and HTTPS work
21   together to ensure that <br>
22   web pages are created, displayed, and communicated securely.
23   <br> As a result, web technologies play a major role in modern communication,
24   business, education, and everyday digital activities.
25 </p>
26
27 <p>
28 <a href="https://www.w3schools.com">Click here</a>
29 </p>
```



Understanding Web Technologies



Web technologies are important because they make it possible for people and organizations to create, access, and share information and services through the internet. They support websites, web applications, online communication, e-commerce, banking, education, and many other digital services. Technologies such as HTML, CSS, JavaScript, HTTP, and HTTPS work together to ensure that web pages are created, displayed, and communicated securely. As a result, web technologies play a major role in modern communication, business, education, and everyday digital activities.

[Click here](#)

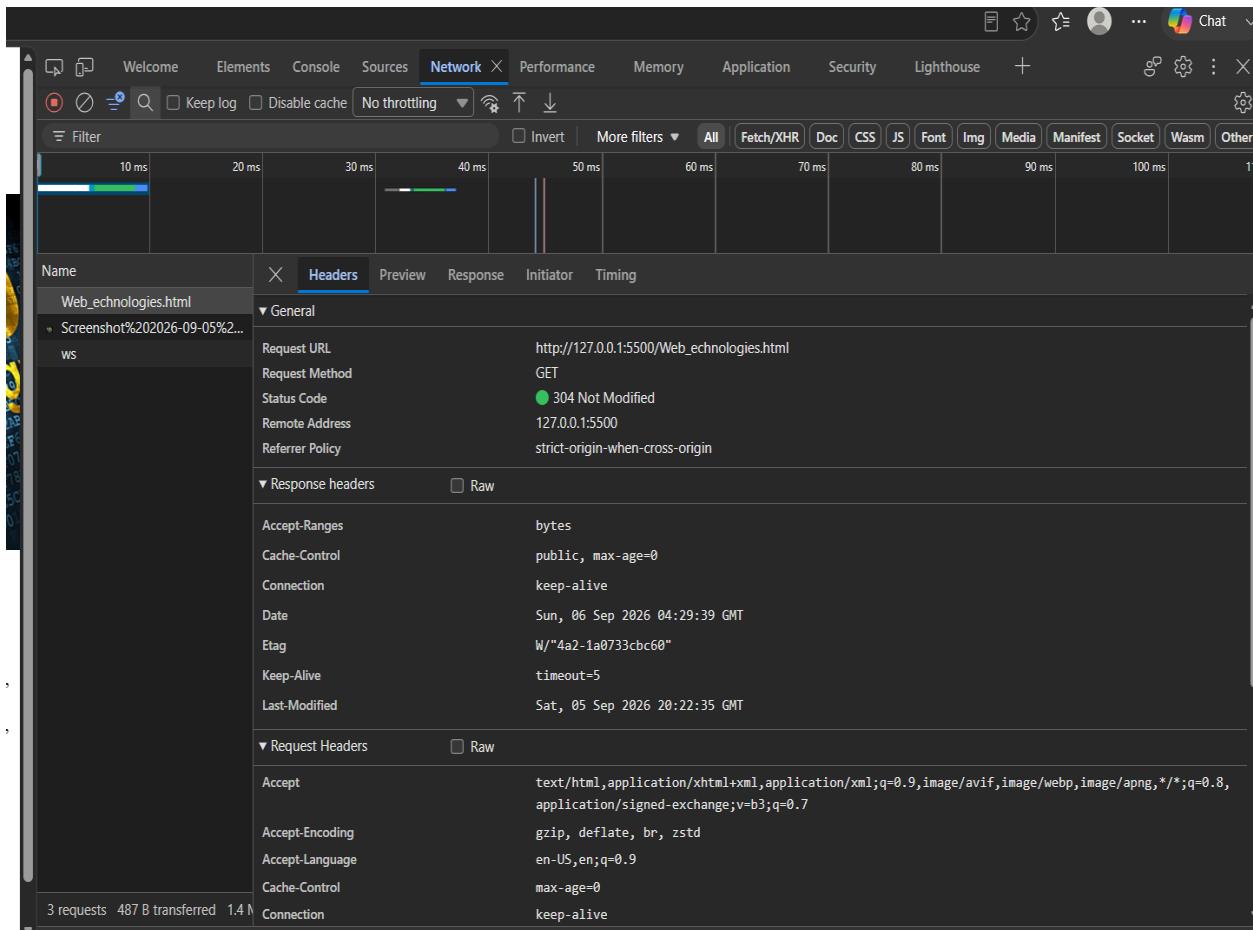
Reflection:

HTML documents have a clear structure that helps organize web content. Headers provide titles, paragraphs present information, and links help users navigate to other pages. Together, they make a webpage easier to read and use.

Exercise 2: Analyzing HTTP Requests and Responses

Command Used to inspect:

F12



Findings :

Item	result	Meaning
Request URL	http://127.0.0.1:5500/Web_echnologies.html	The local address of HTML page
Request Method	GET	The browser is requesting the page

Status Code	304 Not Modified	The page has not changed, so the browser can use its cached version
Cache-Control	public, max-age=0	The response can be cached but should be revalidated
Date	Sun, 06 Sep 2026 04:55:22 GMT	Date/time of the response
Connection	keep-alive	The connection can remain open for further requests
ETag	W/"4a2-1a0733cbc60"	Identifies a particular version of the resource
Last-Modified	Sat, 05 Sep 2026 20:22:35 GMT	Shows when the HTML file was last modified

What does a status code of `200 OK` mean?

Status Code 200 OK means the request was successful and the server successfully returned the requested resource.

Reflection:

HTTP requests and responses are important for communication between a browser and a web server. The request method shows what action is being performed, while the status code shows whether

the request was successful. Response headers provide additional information about the data and how it should be handled.

potential vulnerabilities associated with improperly handled requests and responses?

Understanding these components is important for developing and securing web applications because improperly handled requests can lead to vulnerabilities such as unauthorized access, injection attacks, and data exposure.

Exercise 3: Understanding HTTPS and Its Importance

1. Inspection of HTTPS

The screenshot displays the Chrome DevTools Network tab. The top toolbar includes buttons for 'Filter', 'Invert', 'More filters', and various resource type filters (All, Fetch/XHR, Doc, CSS, JS, Font, Img, Media). A timeline at the top shows a request occurring around 1,000 ms. The main panel shows a list of requests on the left, with the first request selected: a GET request to `https://www.google.com/shared_dict/homepage/66486b7cda479683c8e369fba813822c503ba3ea2c62194e1`. The right panel shows the details for this request, including the 'General' tab with fields for Request URL, Request Method (GET), Status Code (200 OK (from disk cache)), Remote Address (142.251.152.119:443), and Referrer Policy (no-referrer). Below this, the 'Response headers' tab is active, showing headers such as 'Accept-Ranges: bytes', 'Age: 105494', 'Alt-Svc: h3=':443'; ma=2592000, h3-29=':443'; ma=2592000', 'Cache-Control: public, max-age=31536000', 'Content-Encoding: br', 'Content-Length: 54450', and 'Content-Security-Policy: require-trusted-types-for 'script'; report-uri https://csp.withgoogle.com/csp/search-dictionary-policy'.

Name	Request URL	Request Method	Status Code	Remote Address	Referrer Policy
66486b7cda479683c8e369fba...	https://www.google.com/shared_dict/homepage/66486b7cda479683c8e369fba813822c503ba3ea2c62194e1	GET	200 OK (from disk cache)	142.251.152.119:443	no-referrer

Header	Value
Accept-Ranges	bytes
Age	105494
Alt-Svc	h3=':443'; ma=2592000, h3-29=':443'; ma=2592000
Cache-Control	public, max-age=31536000
Content-Encoding	br
Content-Length	54450
Content-Security-Policy	require-trusted-types-for 'script'; report-uri https://csp.withgoogle.com/csp/search-dictionary-policy

Findings:

- Request URL: [https://www.google.com/...](https://www.google.com/)
- Request Method: GET
- Status Code: 200 OK (from disk cache) - the request was successful, and the browser loaded the resource from its local cache.
- Remote Address: 142.251.152.119:443 - Port 443 is the standard port commonly used for HTTPS traffic.
- Cache-Control: public, max-age=31536000 - the resource can be cached for a long period.
- Content-Encoding: br - Brotli compression was used.
- Content-Length: 54450 bytes.
- Content-Security-Policy: A security policy that helps control which resources/scripts the browser is allowed to load.

2. Investigate the security certificate details

Certificate Viewer: *.google.com

GeneralDetails

Issued To

Common Name (CN)	*.google.com
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	WR2
Organization (O)	Google Trust Services
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Monday, August 10, 2026 at 9:37:35 AM
Expires On	Monday, November 2, 2026 at 9:37:34 AM

SHA-256 Fingerprints

Certificate	8f95cc30e88f6b71ef351f710332850c5544e7594af40b1a7e119e18efd61022
Public Key	046f8870e1bd047f0448b7a71f285b399c06d5db52d0283c6333f2a84cefb733

:

- **Certificate issuer:**WR2
- **Validity period:**Monday,August 10,2026 - Monday Nov 2,2026
- **Hash algorithm observed: SHA-256**

2. **Reflection:** HTTPS provides a more secure connection than HTTP. It uses encryption to protect information exchanged between the browser and the web server. This helps prevent attackers from reading sensitive information while it is being transmitted. HTTPS helps protect communications against interception and tampering when certificate validation and the underlying TLS configuration are properly implemented
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WEB TECHNOLOGIES LAB REPORT

Introduction

Web technologies are the tools and methods used to create and operate websites and web applications. In this lab, I learned about HTML, HTTP, and HTTPS and how they work together when a user visits a website. I also used the browser Developer Tools to observe how web pages communicate with servers.

Exercise 1: Understanding HTML Structure

In this exercise, I learned about the basic structure of an HTML document. HTML is used to create the content and structure of a webpage.

I worked with elements such as headings, paragraphs, and links. Headings help organize information, paragraphs are used for text, and links allow users to move from one page or resource to another.

Reflection

This exercise helped me understand how the different parts of an HTML document work together. I learned that using a proper structure makes a webpage easier to read, use, and maintain.

Exercise 2: Analyzing HTTP Requests and Responses

I used the browser Developer Tools and opened the Network tab to observe the communication between my browser and the local web server. After refreshing the page, I selected my HTML file to view its request and response information.

Findings

Request URL:

`http://127.0.0.1:5500/Web_echnologies.html`

Request Method:

`GET`

Status Code:

`304 Not Modified`

Cache-Control:

`public, max-age=0`

Date:

`Sun, 06 Sep 2026 04:55:22 GMT`

Connection:

`keep-alive`

ETag:

`W/"4a2-1a0733cbc60"`

Last-Modified:

`Sat, 05 Sep 2026 20:22:35 GMT`

Explanation

The GET method means that the browser was requesting the HTML page from the server.

The 304 Not Modified status means that the page had not changed since the previous version stored by the browser. Therefore, the browser could use the cached version instead of downloading the complete file again.

The Cache-Control header provides instructions about how the browser should handle the cached resource. The ETag and Last-Modified headers help the browser determine whether the file has changed.

Reflection

This exercise helped me understand what happens in the background when a webpage is opened. Before this exercise, I mainly saw the webpage itself, but I did not pay much attention to the communication taking place between the browser and server.

I also learned that HTTP requests and responses need to be handled properly because weaknesses in web communication can lead to problems such as data exposure, injection attacks, and unauthorized access.

Screenshot: Network request and response headers from my HTML page.

Exercise 3: Understanding HTTPS and Its Importance

For this exercise, I visited a website using HTTPS and used Developer Tools to inspect its network activity.

HTTPS Request Findings

Request Method:

GET

Status Code:

200 OK (from disk cache)

Remote Address:

142.251.152.119:443

Cache-Control:

public, max-age=31536000

Content-Encoding:

br

I noticed that the connection was using port 443, which is commonly used for HTTPS traffic. The 200 OK status showed that the request was successful.

I also observed a Content-Security-Policy response header. This provides additional protection by controlling certain types of content that a browser is allowed to load.

Certificate Details

Certificate Issuer:

WR2

Validity Period:

Monday, August 10, 2026 – Monday, November 2, 2026

Cryptographic Hash:

SHA-256

Reflection

This exercise helped me understand why HTTPS is important when using websites. HTTPS protects information while it is being transferred between the browser and server. This makes it harder for attackers to read or change the information during transmission.

I also learned that websites using HTTPS have digital certificates that provide information about the website's security and validity. The certificate has an issuer and a validity period. SHA-256 is a cryptographic hash algorithm used as part of certificate and digital signature security.

HTTPS is especially important when entering sensitive information such as passwords, payment details, or personal information.

Screenshot: HTTPS network request and response headers.

Screenshot: Website security certificate details.

Recommendations

I recommend that:

- Websites should use HTTPS instead of unprotected HTTP.
- Developers should properly validate user input.
- Sensitive information should be protected during transmission.
- Security headers should be properly configured.
- Web servers and software should be kept updated.

- Developers should regularly inspect and test their applications for security weaknesses.
- Caching should be configured carefully, especially when dealing with sensitive information.

Conclusion

In conclusion, web technologies are an important part of modern websites and applications. HTML is used to structure webpage content, HTTP allows communication between browsers and servers, and HTTPS provides protection for that communication.

The exercises helped me understand how web requests work and why secure communication is important. This knowledge is useful not only for web development but also for cybersecurity, where understanding how web applications communicate can help in identifying and preventing security problems.